

Digital Filter Design Using MATLAB

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Digital filters are commonplace and an essential element of everyday electronics such as radios, cellphones and stereo receivers.

A digital filter is a system or network that selectively changes the wave shape, amplitude-frequency or phase-frequency characteristics of a signal in a desired manner. Common filtering objectives include improving the signal quality and extracting information from the signals. A digital filter is a mathematical algorithm implemented in hardware and software that operates on a digital input signal to produce a digital output signal in order to achieve the filtering objective.

Presented here is a MATLAB program for designing digital filters using various techniques.

Types of digital filters

Digital filters often operate on digitised analogue signals or just numbers representing some variables stored in a

computer memory. These are broadly classified into infinite impulse response (IIR) and finite impulse response (FIR). Each type of filter can be represented by its impulse response sequence, $h(k)$ [$k = 0, 1, \dots$]. The input and output signals are related by the convolution

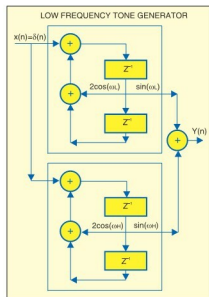


Fig. 1: DTMF generator using two IIR filters connected in parallel

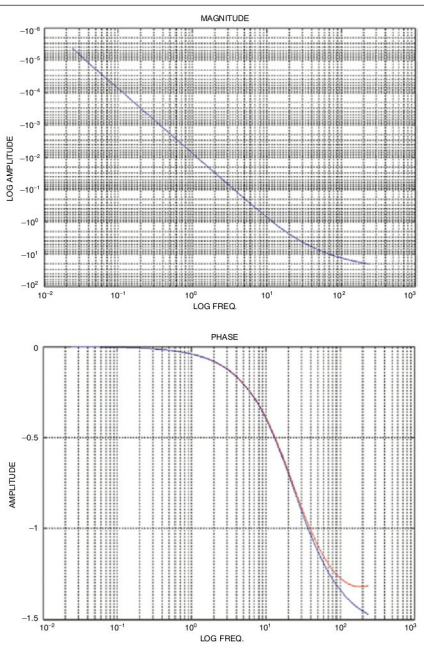


Fig. 2: IIR filters showing magnitude and phase plots